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The value of collateral in trade finance☆

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ABSTRACT

Suppliers are subject to the credit risk of their customers when they sell products on credit. However, rights to the collateral value of the products they sell may mitigate some of this risk. This paper demonstrates the important role of laws that support suppliers' rights to reclaim and liquidate collateral. Using a change in the US bankruptcy code that altered the rights of a subset of suppliers, I use a difference-in-differences setting to show that an improvement in suppliers' rights to the liquidation value of collateral results in an increase in the amount and duration of trade credit offered. The increase in collateral protection also reduced suppliers' lending standards, resulting in more dispersed trade credit lending and riskier customer portfolios. Finally, I find that the increase in collateral rights decreased suppliers' incentives to monitor their customers, consistent with collateral and monitoring being substitutes. Overall, the paper shows that with strong legal protections in place, trade credit has an important collateral component.

1. Introduction

Trade credit is the largest form of short-term, external finance for firms in the United States. Up to 90% of inter-firm trade is supported by trade finance (Auboin, 2009), and accounts payable represent one of the largest short-term liabilities on the balance sheet of US corporations. While selling on credit comes with many advantages like higher order volume, it is not without potential costs. Specifically, companies selling on credit increase their exposure to credit risk spillovers if their customers fail to repay them. This paper investigates the role of creditor protections in supporting inter-firm trade. Specifically, I ask whether legislation that improves trade creditors' rights to collateral impacts their lending behavior.

Much of the prior literature on trade credit ponders the question of why companies extend credit in the presence of an established banking industry. For example, why would a supplier borrow short term from a bank and subsequently lend short term to his customer? One prominent theory explaining the use of trade credit suggests that, relative to a bank, the supplier has an advantage in his ability to liquidate the goods sold to the customer on credit. Because the original seller is better positioned to deal with the collateral than an outside investor, the trade creditor

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may be willing to lend at a cheaper rate than the bank otherwise would (Longhofer and Santos, 2003; Frank and Maksimovic, 1998). In other words, suppliers may be more willing to extend credit in circumstances where collateral liquidation values are higher.

Critical to the collateral theory of trade credit is the idea that trade creditors have the legal right to the underlying collateral. However, products that have already been transformed or sold are difficult to reclaim, and the existence of higher priority lenders could limit trade creditors' recoveries. Therefore, exploring the role of collateral in trade credit requires an understanding of the legal institutions supporting inter-firm trade.

To provide evidence on the link between collateral values and trade credit, I study a change in the US bankruptcy code that significantly altered trade creditors' rights in Chapter 11. The Bankruptcy Abuse Prevention and Consumer Protection Act of 2005 (BAPCPA) dramatically shifted the Chapter 11 landscape for trade creditors. While the primary focus of the law was to make it more difficult for individuals to file bankruptcy, a few provisions in the law also impacted corporations. First, Section 546(c) extends the reclamation reach-back period to 45 days after the buyer's receipt of goods. Relative to the pre-BAPCPA reclamation period of ten days, 546(c) allowed the supplier a significantly longer window to reclaim the goods from sales leading up to the bankruptcy. The second provision is Section 503(b)(9), which bumps the supplier up from an unsecured creditor to an administrative claimant for the amount of the goods sold—this significantly improves the trade creditor's chances of securing the full cash value of his collateral. Ceteris paribus, both 503(b)(9) and 546(c) should improve the supplier's ability to recover the value of the goods in the event of a customer's bankruptcy.

BAPCPA is a well-suited setting to study the role of collateral in the trade credit decision for a number of reasons. First, the law represents a distinct improvement in the supplier's legal rights to the value of collateral. Prior to the enactment of BAPCPA, trade creditors relied on recoveries through relatively weaker state reclamation laws governed by the Uniform Commercial Code (UCC). Under the UCC laws, the reclamation period was limited to ten days, and the trade creditor had no rights to administrative expense claims. As a result, BAPCPA represents a large shift in the potential recoveries for trade creditors in Chapter 11.[1] Second, whereas prior studies rely on cross-country variation in lender rights to investigate how collateral value impacts lending, BAPCPA represents a change in lenders' rights to collateral within the United States. This helps to rule out alternative explanations for observed lending behavior such as variation in enforcement standards and economic conditions. Finally, BAPCPA impacted only a subset of suppliers in the US, since reclamation and administrative expense rights only apply to a certain set of goods producers rather than service producers and commodities suppliers. This allows for investigation of a treatment effect benchmarked against a non-treatment control sample.

One challenge in measuring how collateral values impact trade credit is that such analysis requires detailed trade data that is granular enough to capture time-series variation in the amount of credit extended and the terms of trade while also identifying the types of products or services provided. To overcome this challenge, I rely on two unique and detailed trade data sets. First, I employ a unique data set of inter-firm supply contracts collected from SEC filings. The contracts provide details of material trade agreements filed by public companies, including identities of the trading partners, product specifications, and the duration of the trade credit offered.[2] Second, I use a proprietary database of detailed trade credit transactions between buyers and sellers, provided by a large credit scoring company. The data not only provides the identity of buyer-supplier pairs, but it also identifies the volume of trade credit extended for every transaction executed between a buyer and a seller on a monthly basis. Notably, the granularity of the data allows me to include buyer-month fixed effects to control for variation in the demand for credit, thus helping to isolate the role of collateral in the supply of trade credit. Together, the data sets provide information on the duration and the volume of trade credit as well as product characteristics that allow me to classify firms as treated suppliers under BAPCPA rules.

Critical to my empirical approach is the assumption that BAPCPA led to a higher expected recovery rate for trade creditors that sell goods. Consistent with this assumption, I provide evidence that BAPCPA indeed improved trade creditor recoveries after the passage of the law. Suppliers' account receivable write-offs, which typically occur when the firm expects that they will not recover the loan, declined by 9% following the passage of BAPCPA. Further, suppliers were 11% less likely to be listed in the lowest priority category of unsecured creditors in bankruptcy filings following BAPCPA. Importantly, the changes in recovery rates only impacted the treated suppliers—those that supplied goods as defined by BAPCPA. The evidence suggests that BAPCPA was successful in increasing the liquidation value of collateral for a subset of suppliers.

To capture whether a change in collateral values alter trade creditor behavior, I use a difference-in-differences approach. I categorize goods producers that meet the requirements of BAPCPA 503(b)(9) and 546(c) as treated suppliers, and all other suppliers are included in the control sample. Following the theoretical models linking collateral value to the decision to extend trade credit (Longhofer and Santos, 2003; Frank and Maksimovic, 1998), I predict that the suppliers in the treatment sample will show a greater increase in both the amount and duration of trade credit that they offer to customers after the enactment of BAPCPA, relative to the control sample. Based on the difference-in-differences test, I find that the treated suppliers increased the trade credit payment due date by ten days more than the control sample. Further, I find that

[1] http://www.jdsupra.com/legalnews/the-chapter-11-vendor-game-changer-87882/.

[2] The duration of trade credit is the number of days after invoicing that the customer has to pay the bill. The sample of contracts state the ex-ante agreed upon trade credit days for each delivery covered throughout the contractual period.

they increase the volume of trade credit extended about 40% more than the control sample. The evidence is consistent with models predicting a causal link between the liquidation value of collateral and the extension of trade credit.

Which customers get more trade credit? One possibility is that suppliers increase credit equally to all customers in their portfolio. However, another possibility is that higher recovery rates allow suppliers to spread their lending pool to less creditworthy customers. For example, Mian and Sufi (2009) find that mortgage securitization allowed banks to lend to riskier homeowners. I find that, following BAPCPA, treated suppliers increase the amount and duration of credit on both the intensive and extensive margins. The evidence suggests that treated suppliers offer credit to a larger number of customers, and the concentration of their lending pool declines more after BAPCPA relative to untreated suppliers. To provide evidence on the mix of customers receiving credit, I match the customers to indicators of their credit risk and show that after BAPCPA, treated suppliers hold a riskier portfolio of receivables; the weighted average credit score of their customer portfolio declines by 7% more than the control sample's customer portfolio. The evidence is consistent with the notion that increased protection from collateral enhances the lender's incentive to bear additional risk.

Next, I investigate whether BAPCPA enhanced or reduced the supplier's incentive to monitor. Manove et al. (2001) model the link between creditor rights and monitoring incentives and find that collateral and screening are substitutes; the protection offered from collateral decreases the investor's incentive to monitor. On the contrary, Rajan and Winton (1995) model how collateral might improve lenders' incentives to monitor. When the lender's payoff is sensitive to the borrower's financial health, collateral is only effective if the lender can monitor its value. Therefore, whether BAPCPA increased or reduced the trade creditor's incentives to monitor is an open empirical question. To provide evidence on this relation, I investigate the frequency with which suppliers check the credit scores of their customers. I find that treated suppliers decrease customer credit checks about 50% more than untreated suppliers after BAPCPA. The negative relation between collateral and monitoring suggests that, in spite of the supplier's incentives to increase customer portfolio risk, collateral reduces the returns to monitoring.

Finally, I investigate how the shift in collateral rights impacts other lenders. By increasing the rights of trade creditors, other creditors (i.e., banks) may face weaker protections in bankruptcy and reduce their lending ex-ante. Using detailed data on firms' debt capital structure, I find that lower-priority, unsecured creditors reduce the credit extended to the treatment group more after BAPCPA, relative to the their lending to the control group. This evidence suggests that while the law improved the collateral rights of some creditors, there may have been some negative spillover effects on the recovery rates and corresponding lending behavior of other creditors.

My paper makes several contributions to the extant literature. First, I contribute to the growing literature that aims to understand the provision of trade credit and its economic consequences. One theoretical explanation for the use of trade credit suggests that suppliers are better positioned to deal with collateral than other lenders, so they should be willing to provide more credit when liquidation values are higher. My paper shows that a key ingredient in the collateral theory of trade credit is that the trade creditor has the legal right to the value of the goods supporting trade. As an important caveat, this paper does not suggest that trade credit is solely, or even primarily, a collateral-based form of lending. Rather, the findings highlight that with strong legislation in place that supports the trade creditors' rights in bankruptcy, trade credit can have an important collateral component.

More generally, this paper is related to the large literature investigating how the legal system impacts lending markets. Prior literature shows that cross-country variation in creditor protection is linked to the size of credit markets (La Porta, Lopez-de Silanes, Shleifer and Vishny, 1997; 1998; Levine, 1998; 1999; Djankov, McLiesh and Shleifer, 2007). These prior studies suggest that stronger laws allow creditors to enforce contracts with their borrowers, thus reducing the cost of external financing. Within this broad literature, my paper is closely related to studies showing that collateral rights are an important mechanism through which legal protections impact financing (Benmelech and Bergman, 2009; Cerqueiro, Ongena and Roszbach, 2016; Sautner and Vladimirov, 2018). My research setting allows for plausibly exogenous variation in collateral values for a subset of creditors within the US, which helps to isolate the importance of collateral protection as an explanation for observed lending behavior. I show that improving collateral protection for a subset of lenders increases their willingness to lend. However, shifting collateral to trade creditors reduces the protections of other classes of lenders, who subsequently decrease their lending. Thus, laws like BAPCPA that target the rights of some creditors can have important policy implications, depending on whether they achieve a more efficient or a more suboptimal allocation of rights among lenders.

Finally, the paper illustrates how collateral rights relate to screening and monitoring standards. Specifically, improving suppliers' legal rights to collateral results in riskier customer portfolios and reduces incentives to monitor the borrower. In this way, the paper speaks to the important literature that studies the relation between securitization and lending and monitoring standards (Manove, Padilla and Pagano, 2001; Keys, Mukherjee, Seru and Vig, 2010; Mian and Sufi, 2009; Nadauld and Sherlund, 2013) and could have implications for aggregate supply chain risk.

The paper proceeds as follows, Section 2 outlines the framework I rely on for my main predictions, Section 3 describes the research setting, Section 4 discusses my empirical approach, Section 5 describes some notable features of the data, Section 6 presents the main results, Section 7 provides robustness analyses, and Section 8 concludes.

2. Trade credit: theories and evidence

Extending trade credit increases the supplier's exposure to their customers' credit risk, and prior literature shows that the opportunity cost of offering credit is high (Murfin and Njoroge, 2014). As a result, researchers question why suppliers engage in lending when more specialized monitors like banks could fill that role. One prominent theory explaining trade credit suggests that it helps to alleviate frictions between customers and external financiers.[3]

The financing theory posits that trade credit substitutes for bank credit when trade partners face frictions in external financial markets. Prior studies show that trade credit can substitute for bank credit during periods of tight credit or financial crises (Nilsen, 2002; Choi and Kim, 2005; Love, Preve and Sarria-Allende, 2007). Petersen and Rajan (1997) use the National Survey of Small Business Finance (NSSBF) to show that firms with better access to bank credit have higher levels of accounts receivable. The main prediction of the financing theory is that large, creditworthy suppliers will extend credit to customers that have positive net present value projects but are rationed from direct credit markets.

There are several reasons why suppliers might be willing to lend when banks are not. First, trade creditors may place a higher collateral value on borrowers' assets. Longhofer and Santos (2003) and Frank and Maksimovic (1998) model the decision to extend trade credit with the assumption that the supplier has an advantage in valuing the underlying assets, particularly when those assets are not homogenous. Because suppliers have a network of alternative buyers, they place a higher liquidation value on the assets and will lend at a cheaper rate than banks otherwise would. Key to the collateral theory of trade credit is the assumption that suppliers have strong legal rights and institutions supporting their ability to reclaim and/or liquidate collateral.

A similar stream of literature hypothesizes that suppliers have an informational advantage over banks, enabling them to screen and monitor loans more effectively. Smith (1987) and Biais and Gollier (1997) argue that suppliers have an informational advantage over arm's length financiers because of the frequency of interactions and the types of information exchanged. For example, the supplier may visit the buyer's premises more often than financial institutions or may obtain information about the buyer's creditworthiness through demand forecasts and other operational information. Burkart and Ellingsen (2004) model the trade credit decision and note that it is less profitable for an opportunistic borrower to divert inputs rather than to divert cash; this reduces the monitoring costs of the trade creditor, putting him at an informational advantage relative to banks.

Finally, suppliers may be willing to finance their customers because their payoff differs from that of the bank. Wilner (2000) develops a model showing that trade creditors, desiring to maintain an enduring product market relationship, grant more concessions to a customer in financial distress than would be granted by lenders in competitive lending markets. Similarly, Cunat (2007) argues that if the trade relationship involves specific investments, customers have less incentive to default on their suppliers than on their banks, and suppliers have stronger incentives to extend credit to distressed buyers.

While some of the empirical literature shows a link between product characteristics and trade credit, there is no evidence thus far that causally links collateral values to the trade decision. For example, Giannetti et al. (2011) note that the documented link between product characteristics and trade credit could be explained by high switching costs (Cunat, 2007), moral hazard (Burkart and Ellingsen, 2004), information problems (Smith, 1987), or collateral values. The collateral theory, while linked to product characteristics, relies on two critical assumptions: (1) suppliers have the legal right to repossess or liquidate goods from a non-paying customer, and (2) the liquidation value of the collateral determines the amount that a supplier is willing to lend. By using BAPCPA as a research setting, I exploit a large shift in both the repossession rights and liquidation values of the goods supplied.

3. The Bankruptcy Abuse Prevention and Consumer Protection Act of 2005

The Bankruptcy Abuse Prevention and Consumer Protection Act of 2005 (BAPCPA) made several changes to the US bankruptcy code. Most of the changes in the code affected individuals filing bankruptcy. Principally, the law made it more difficult for consumers to file for Chapter 7 bankruptcy. Debtors whose monthly income is higher than the median income of their state are subject to a "means test" to determine whether there is abuse and therefore there are grounds for dismissal.

What brought about BAPCPA? A spike in bankruptcy filings in the 1990s gave rise to arguments that the bankruptcy code was too debtor friendly and allowed for bankruptcy abuse and fraud. Credit card companies initiated the changes by lobbying Congress, arguing that enforcing the obligation to pay would promote credit availability to individuals and reduce interest rates. Critics argued that the bill would impose disproportionate costs on non-opportunistic debtors who need access to bankruptcy relief. To inform the debate, Congress formed the Bankruptcy Review Commission to conduct hearings about the current bankruptcy system and to propose solutions. Interestingly, the commission did not recommend changes to corporate bankruptcy laws impacting the suppliers of goods.[4]

After a five-year lobbying effort, BAPCPA was introduced on February 1, 2005. It was signed into law by President George Bush on April 20, 2005, though most provisions in the act apply to cases filed on or after October 17, 2005.

[3] The literature also identifies other, non-financial, incentives for suppliers to lend. For example, Long et al. (1993), Lee and Stowe (1993), and Kim and Shin (2012) suggest that suppliers allow delayed payment to let customers check the product quality. These models suggest that trade credit can be used as a commitment device to mitigate moral hazard problems in production chains. Further, Klapper et al. (2011) and Fabbri and Klapper (2016) show that trade credit terms are the result of the party's relative bargaining power. Finally, Antras and Foley (2015) show that in an international setting, country-level enforcement standards impact trade credit terms.

[4] http://govinfo.library.unt.edu/nbrc/report/01title.pdf.

3.1. Pre-BAPCPA trade creditor

Prior to BAPCPA, suppliers to a debtor in the prepetition period were considered general unsecured claimants, without any guarantee that their claims be paid in full. Anecdotal evidence suggests that the recovery rates of unsecured creditors are historically low.[5]

One option for suppliers to increase recoveries is to reclaim the goods from bankrupt customers. Prior to BAPCPA, reclamation rights were governed by Article 2-702 of the UCC.[6] The UCC allows a vendor who sold goods on credit to a bankrupt customer to file for reclamation rights to those goods—as long as a written request is filed within ten days of receipt of those goods. However, the actual recoveries under reclamation were limited because (1) secured lender rights typically trumped the reclamation rights of the supplier; (2) the ten-day reach-back period was often considered too short for many smaller suppliers to file, and it disqualified any goods that were delivered to the customer more than ten days preceding the bankruptcy; and (3) if the goods were sold or transformed by the debtor before the reclamation demand was made, the vendor's reclamation rights were lost.

3.2. Post-BAPCPA trade creditor

First, BAPCPA Section 546(c) enhanced the reclamation rights of the trade creditor by extending the reach-back period to 45 days. In other words, a trade creditor now has the ability to reclaim up to 45 days' worth of deliveries that precede the customer's bankruptcy, as opposed to only ten days in the pre-BAPCPA regime. To make a 546(c) claim, the seller must establish that (1) the seller provided goods sold in the ordinary course of business; (2) the debtor received such goods while insolvent, within 45 days before commencement of the case; and (3) the seller made a written demand for reclamation within 45 days.[7]

Second, Section 503(b)(9) added an administrative expense priority for the value of any goods received by the debtor within 20 days prior to the commencement of the case. The claim is allowed for the value of any goods received by the debtor in the ordinary course of business within 20 days before the date of commencement of a case."[8] Congress added 503(b)(9) to the Bankruptcy Code as part of Section 1227 of BAPCPA, entitled "Reclamation," and it is often used as a safety net for 546(c) in the case that the seller has failed to give adequate notice for reclamation. Since administrative expense claims must be paid upon confirmation of a plan, the status of the trade creditor was boosted from a general unsecured claimant, often recovering only a small portion of the value of the goods, to a higher secured lender, often recovering 100% of the value of the collateral.[9]

It is not clear what precipitated the addition of these laws, as the legislative history of the adoption of 503(b)(9) and 546(c) is non-existent.[10] Though no particular group lobbied for 503(b)(9) or 546(c), the sections are consistent with the spirit of the law that advanced certain creditor interests.[11] Further, legal scholars have noted significant latitude in the court's interpretation of both provisions. In particular, the court needs to assess the "value" of the vendor's claim, whether the items can be classified as "goods," and whether the items were sold in "the ordinary course of business." Initial bankruptcies filed under BAPCPA reflected some variation in the court's interpretation of 503(b)(9). Therefore, while BAPCPA laws reflected dramatic changes for the trade creditor, whether those changes materialized is an empirical question.

3.3. BAPCPA and trade creditors' assessment of collateral value

I argue that BAPCPA impacts trade credit extended through an improvement in suppliers' expectations about the liquidation value of the collateral. The trade creditor cares about the financial value that he receives in the customer's default state, thus an improvement in the recovery value of the collateral should improve his willingness to lend ex-ante. 546(c) reduces the costs of reclaiming goods, so ceteris paribus should improve the supplier's recovery value.

503(b)(9) has a more nuanced link to collateral value since it elevates the priority status of the trade creditor. In a trade credit loan, an improvement in priority typically represents a simultaneous improvement in collateral liquidation value. This is because the supplier lends goods, not cash, and is owed the value of those goods in return. Thus, a provision that improves the priority of the trade creditor's loan simultaneously improves the liquidation value of the collateral.[12] The link between 503(b)(9) and collateral liquidation value is also implicit in the provision's positioning in the law (i.e., under Section 1227 entitled "Reclamation") as well as in the language of 503(b)(9) which specifies that a claim can be made for "...*the value of any goods* received by the debtor..." [emphasis added].[13]

Taken together, my assumption is that trade creditors perceive both 503(b)(9) and 546(c) as improving

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5 https://www.allmandlaw.com/Blog/2010/May/Unsecured-Creditors-Desperately-Fight-For-Pennie.aspx.

6 UCC 2-702 is state law in all U.S. jurisdictions except Louisiana.

7 https://www.gpo.gov/fdsys/pkg/PLAW-109publ8/html/PLAW-109publ8.htm.

8 http://503b9.com.

9 503(b)(9) administrative claims can be made with very few conditions: (1) there is no need to prove the debtor's insolvency; (2) there's no need to establish the seller's right to the goods under state law; and (3) an administrative claim can be made without any connection to the right of reclamation.

10 Discussions with legal scholars and bankruptcy attorneys suggest that no particular group lobbied for 503(b)(9) or 546(c). See https://www.hunton.com/images/content/3/5/v2/3576/ABI_Journal_Section_503.b.9_2.11.pdf for a discussion of the lack of legislative history on these two sections to the code.

11 I address further concerns about the validity of the BAPCPA setting in Section 7.

12 I assume that the trade creditor does not plan to use the collateral himself but rather values the financial recovery of the value of goods in the case that the buyer defaults. Since, 503(b)(9) directly reduces the costs of liquidating the goods (storing in inventory, possible fall in price due to selling the repossessed goods to the next best user, etc), it increases the supplier's financial recovery of the collateral value.

13 https://www.gpo.gov/fdsys/pkg/PLAW-109publ8/html/PLAW-109publ8.htm.

their collateral liquidation value. However, I recognize that 503(b)(9) is a bundled improvement of both priority and collateral liquidation value, thus it may impact the inferences of the results. I address this further in Section 6.2.

4. Identification strategy

4.1. Treatment effect

Critical to my empirical strategy is the assumption that BAPCPA increased the expected liquidation values for trade creditors. As discussed above, there was some concern in the legal literature that ambiguity in the language of BAPCPA gave courts latitude in enforcement. Indeed, anecdotal evidence suggests that some judges failed to rule in favor of trade creditor recoveries.[14] If trade creditors anticipate that BAPCPA would not materially change their recovery rights, this would reduce my ability to capture the intended treatment effects. To validate the research setting as a shock to trade creditor recoveries, I would ideally compare the recovery rates of the treatment group before and after BAPCPA. Because recovery rates are not observable, I instead investigate how trade creditor write-offs and the percentage of trade debt appearing in the unsecured creditor pool change around BAPCPA.

Accounts receivable are typically "written off" when managers determine an account is uncollectible. For example, if a customer enters bankruptcy and the supplier anticipates non-payment, at that point the supplier would reduce the accounts receivable balance through a write-off.[15] If BAPCPA increased the likelihood that suppliers would receive payment, either through reclamation or an administrative expense claim, then I expect BAPCPA to decrease write-offs. I obtain write-off information from an annual survey of write-off behavior conducted by the Credit Research Foundation (CRF). CRF asks their members (often credit managers) to report their account receivable write-offs as a percentage of sales, and they report the information at suppliers' two-digit SIC level. This allows me to separate write-off behavior into that of goods suppliers (treatment sample) versus service suppliers (control sample). Fig. 1 illustrates the write-off behavior of suppliers over the 2002-2008 period. Consistent with BAPCPA improving collectability for certain trade creditors, goods suppliers have a dramatic reduction in write-offs around the time of treatment; write-offs drop from around 17% of sales in 2004 to 5% of sales in 2006. No such trend exists for suppliers in the control sample.

Second, I validate BAPCPA as a shock to trade creditor recoveries by examining the amount of trade debt appearing in the unsecured creditor pool. Whereas prior to BAPCPA, trade creditors universally fell into the unsecured creditor pool, after BAPCPA trade creditors had the option to file a 503(b)(9) claim to jump to administrative status. Therefore, I would expect trade debtors to represent a lower percentage of total unsecured debt after BAPCPA.[16] Bankrupt debtors are required to list the creditors holding the 20 largest unsecured claims in Chapter 11 cases, including the name of the creditor, the amount of the claim, and the type of claim (i.e., trade debt). I collect unsecured debt claims from bankruptcydata.com over the period 2002-2008 and calculate unsecured trade debt as a percentage of total unsecured credit. I hand-match the trade creditor names to Compustat to obtain industry information and split the trade claims into those for the treatment group (goods providers) and those for the control group. Fig. 2 shows that after BAPCPA, treated suppliers are less likely to have unsecured claims; unsecured trade debt declines from around 30% in 2004 to around 22% in 2006, consistent with trade creditors moving out of the unsecured creditor pool. Unsecured debt for the control sample remains relatively constant throughout the sample period.

4.2. Main specification

The main prediction of the collateral theory of trade credit is that higher liquidation values increase the amount of trade credit suppliers are willing to lend. I investigate suppliers' response to a change in collateral value using a difference-in-differences (DD) approach, which allows me to control for contemporaneous events that impacted all US firms around the time of BAPCPA. Since both 503(b)(9) and 546(c) both stipulate that the changes only apply to goods suppliers, the setting offers a clean analysis of treatment effects while controlling for otherwise similar firms that were not impacted by BAPCPA. My treatment sample therefore only includes those transactions where the suppliers provide "goods" as defined by BAPCPA, while my control sample includes all other suppliers.[17]

The DD approach compares the amount of trade credit offered after BAPCPA to the amount of trade credit offered before BAPCPA, for both treatment and control firms. The DD is estimated by OLS using the following specification:

$$y_{i,j,t} = \beta_1 Post_{i,j,t} + \beta_2 Treatment_i * Post_{i,j,t} + \gamma Controls_i \\ {}^*Post_{i,j,t} + \alpha_i + \alpha_j + \alpha_t + \alpha_{i,j} + \alpha_{t,j} + \varepsilon_{i,j,t}, \qquad (1)$$

where i indexes suppliers and j indexes buyers. *Post* is a dummy variable equal to one for trade transactions occurring after the enactment of Sections 503(b)(9) and 546(c), and *Treatment* is an indicator variable set equal to one for goods suppliers.

My assumption is that treatment and control firms would exhibit similar trade credit trends, absent the treatment. It is possible, however, that treatment and control firms differ along some dimensions, which might be correlated with the outcome variables and would bias the DD

[14] In Re Global Home Products, LLC, creditors attempted to collect 503(b)(9) claims, but the court argued that the Code did not explicitly provide for immediate payment. https://bernsteinlaw.com/publications-list/bapcpa-amendments-to-bankruptcy-code-provide-extra-teeth-for-unsecured-credit-sellers/.

[15] Suppliers may write off accounts before a customer enters bankruptcy if they have sufficient information that a specific customer is unlikely to pay.

[16] Ideally, I would directly investigate administrative claim filings; however, this information is not required to be disclosed by debtors. My assumption is that a reduction in supplier debt appearing in the unsecured claims indicates supplier debt moved to administrative claim status.

[17] Details of how I define goods suppliers are provided in the Online Appendix and discussed briefly in Section 5.4.

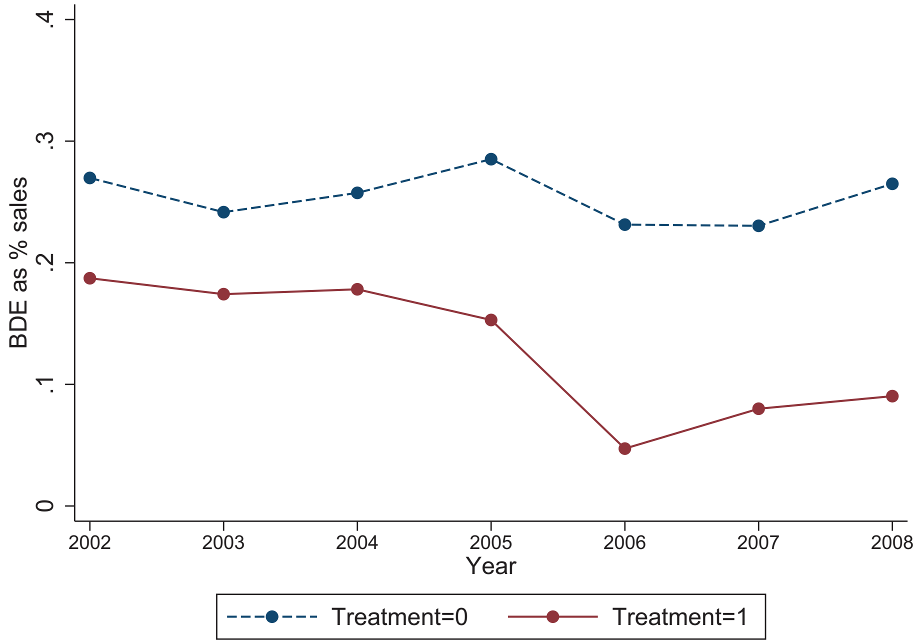

Fig. 1. Bad debt expense as a % of sales over time.
This figure plots account receivable write-offs as a percentage of sales over the 2002–2008 period. Write-offs are aggregated at the supplier's two-digit SIC level. The treatment sample (*Treatment*=1) is represented by the solid line and includes all goods suppliers, as defined in Section 5.4 and in the Online Appendix. The control sample (*Treatment*=0) is represented by the dashed line and includes all supplier industries that do not qualify for 503(b)(9) and 546(c) treatment under BAPCPA.

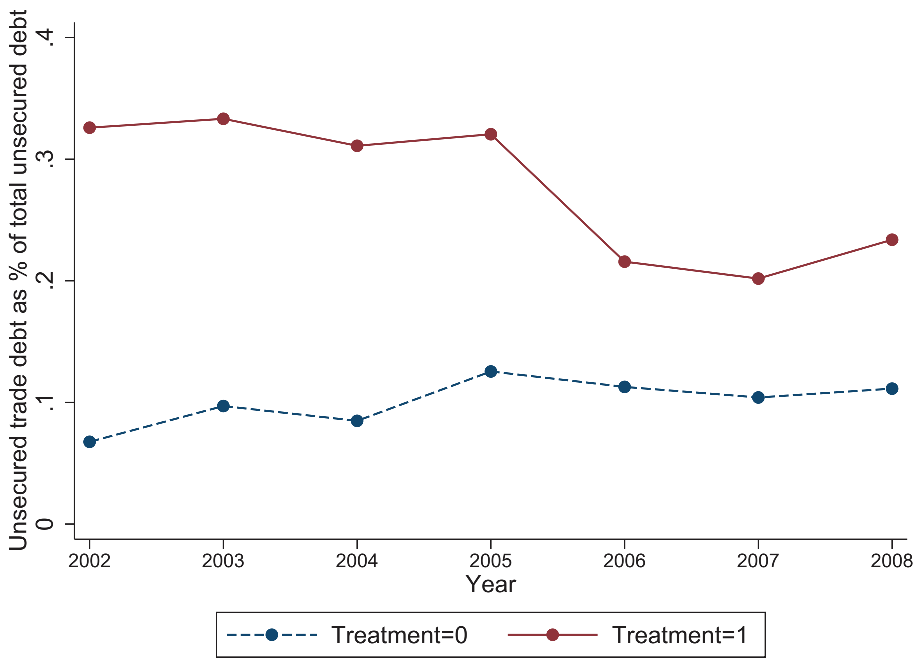

Fig. 2. Unsecured trade claims in bankruptcy as a % of total unsecured claims over time.
This figure plots the volume of unsecured trade credit claims as a percentage of total unsecured claims in bankruptcy. Data is obtained from the list of the 20 largest unsecured claims in Chapter 11 cases, as reported in www.bankruptcydata.com. The treatment sample (*Treatment*=1) is represented by the solid line and includes all goods suppliers, as defined in Section 5.4 and in the Online Appendix. The control sample (*Treatment*=0) is represented by the dashed line and includes all suppliers that do not qualify for 503(b)(9) and 546(c) treatment under BAPCPA.

estimation. To address this issue, I control for firms' pretreatment characteristics as well as their interaction with the *Post* indicator. These controls help to alleviate the concern that my results might be driven by pretreatment variation between the treatment and control groups. They also prevent bias if the treatment and control groups respond differently to macroeconomic fluctuations. The controls do not enter into the regression separately because they are absorbed by the fixed effects. All controls are measured by taking their average over the pre period. Specifically, I control for size (measured by the log of total assets), liquidity (measured by cash and cash equivalents over total assets), firm age (measured as the number of years since a supplier was first observed in Compustat), and leverage (measured by long-term debt over total assets). The controls follow Petersen and Rajan (1997) and Barrot (2016) and are meant to capture variation in firm-specific, observable characteristics that might be correlated with selection into the treatment. All controls are measured by taking the averages over the period 2002 through 2004.[18]

I include an exhaustive set of fixed effects to capture time-invariant characteristics of the supplier and buyer as well as bilateral supplier-buyer relationships. Where possible, I also include buyer-time fixed effects to absorb buyer-specific idiosyncratic shocks and the associated changes in the demand for trade credit. Finally, the main specifications include indicators for the preperiod (the period prior to the beginning of the treatment period) interacted with the *Treatment* indicator to assess any pretreatment trends. The fixed effects, in concert with the identifying assumption of orthogonality of treatment, provide a strong basis to draw causal inference.

5. Data

Data on inter-firm financing is scarce since trade transactions are typically recorded in the internal accounting records of the firm. To study trade credit, most of the prior literature relies on aggregate accounts receivable or accounts payable data reported on balance sheets.[19] While informative, aggregate balance sheet data has three significant drawbacks that limit the ability to capture causal effects: (1) only one side of the transaction is observable; upstream lenders and downstream borrowers are generally unobservable; (2) information about the product or service that accompanies the transaction is unobservable; and (3) quarterly balance sheet data lacks the granularity to assess sharp treatment effects and to identify other features of the trade credit contract like the duration of trade credit offered.

To overcome these deficiencies, I rely on two data sets to obtain trade credit information. The first is a data set of long-term supply contracts that detail the terms of trade including the duration of the trade credit, and the second is a data set of inter-firm sales transactions and the volume of credit extended. While the prior literature typically does not distinguish between different features of the trade credit contract (i.e., duration, volume, price, etc.), it is reasonable to assume that suppliers react to BAPCPA by adjusting multiple trade credit terms.[20] For example, since 503(b)(9) reduces suppliers' losses given default, one would expect that it increases the volume of credit offered. Further, 546(c) increases the number of days of the reach-back period for reclamation. This means that, relative to before BAPCPA, credit sales can be outstanding for a longer period and still qualify for reclamation. Therefore, I would expect that 546(c) increased the duration of trade credit offered.

5.1. Ex-ante contracts: trade credit duration

I use a novel data set of long-term supply contracts collected from SEC filings (the "contracts sample"). Regulation S-K of the Securities Act of 1933 mandates that all publicly filing companies disclose material contracts as exhibits in SEC filings. Included in this requirement is "[any] contract upon which the registrant's business is substantially dependent, as in the case of continuing contracts to sell the major part of the registrant's products or services or to purchase the major part of registrant's requirements of goods, services, or raw materials" (Section 10 (ii)(b) of Regulation S-K).

The sample includes all long-term supply contracts entered into between January 2002 and December 2008.[21] I search SEC filings for exhibits with "supply" or "procurement" in the title and "buyer" and "supplier" or "seller" in the first paragraph, and I search for trade credit details in the body of the contract. The resulting sample that has available trade credit data includes 515 contracts. For each supply contract, I determine whether the filer is the buyer or the supplier of the good or service and match the filing party to Compustat using the Central Indexing Key provided in the filing. Next, I identify the non-filing counterparty by hand-matching the name and location to the Capital IQ database. If the counterparty is public, I obtain relevant financial variables from Compustat (US firms) or Thomson Reuter's Datastream (international firms). The resulting number of contracts with available financial data is 432. Finally, from each contract I extract the product descriptions and the trade credit terms. Since these are ex-ante agreed upon terms of trade, the trade credit terms apply to any transaction falling within the contractual period and are most often expressed as

[18] To further address this issue, I check that all results are robust to matching each supplier in the treatment group with a non-treatment firm within the same quartile of firm size, age, liquidity, and leverage. This procedure helps to further ensure that my results are not driven by pretreatment differences in the control and treatment samples. All results are robust to this matching procedure (untabulated).

[19] Klapper et al. (2011) and Antras and Foley (2015) are notable exceptions. Klapper et al. (2011) investigate trade credit contracts for a sample of 56 large buyers. Antras and Foley (2015) obtain trade credit data from one large poultry exporter.

[20] Unfortunately, I cannot observe trade credit interest rates since they are implicit in discounts offered. Trade credit discounts are not prevalent enough to examine separately.

[21] A subset of this data was used in Costello (2013).

the number of days after an invoice that the payment is due.[22]

5.2. Sales transactions: trade credit volume

Second, I rely on a novel and proprietary data set that contains disaggregated data on accounts receivable for supplier-buyer pairs (the "trade sample"). These data are obtained from Credit2B, a third party credit information platform and reporting service that relies on real-time industry trade reporting. The company produces risk scores for millions of customers worldwide based on credit bureau data, financial filings, news, and trade payment data collected from their member firms. To become a member firm, suppliers are required to produce a monthly report disclosing all transactions with customers; the report contains the current receivables and past due balances for their universe of buyers. In other words, all members must make a monthly upload of every transaction made with every customer.[23] The data also includes a receivables aging report for each transaction. Past due balances are reported for each customer broken into 30-day buckets. Importantly, the information is only used by the management team at Credit2B to produce a risk score for the customer, which is then shared with other member firms. Identities of buyer-supplier pairs and their transaction balances are not disclosed to other member firms, mitigating the concern that self-selection due to reporting incentives might bias the sample.

The sample period for the trade transactions is somewhat shorter than the contracts sample due to data availability. Trade transactions data become reliably populated starting in May of 2004, and thus to have a relatively balanced pre- and post-period, I cut the sample at the end of 2006. The total number of transactions covered over this period is 6.4 million. For the empirical analyses, I also control for firm characteristics including size, liquidity, age, and leverage. Therefore, I match the suppliers in the trade sample to Compustat. To do so, I fuzzy match based on the firm name and location (where possible). The resulting data set includes 514,000 transactions.

5.3. Descriptive statistics

Table 1 reports the summary statistics for the contracts sample (Panel A) and for the trade sample (Panel B). The main dependent variables, *Payment days* and *Current receivable*, reveal that, on average, suppliers extend $31,000 in trade credit per transaction and allow their customers to pay 29 days after invoicing. Of note is the skewness in the current receivables; thus, I take the log of current receivables for the regression analysis. The suppliers in both the contracts sample and the trade sample are well established; the average age of firms in the contracts sample is 19 years, and the average age of the firms in the trade sample is 31 years. To be included in the contracts sample, the trade relationship has to meet a relatively high materiality threshold, and the firms have to have public filings.[24] Therefore, across the sample period there is a modest number of unique suppliers (169) filing contracts with a relatively larger set of unique customers (312). In contrast, the trade sample of Credit2B member firms does not have any materiality thresholds, nor are the firms required to be public. Therefore, the coverage is larger with 1000 unique suppliers over the sample period and close to one million unique customers.[25] The relatively large coverage of customers means that many suppliers share the same customer in a given month, allowing me to control for time-varying customer characteristics in the regression analyses.

To get a sense of potential issues of selection into the sample, I provide the sample averages for the Compustat universe during the period of the experiment. The suppliers in the contracts sample are relatively smaller and younger than the Compustat mean, but they have similar cash holdings and leverage. The trade sample is remarkably similar to the Compustat firms across all dimensions except for firm age. On balance, the suppliers in both samples do not seem wholly different from the Compustat universe.[26]

5.4. Treatment versus control firms

Key to my empirical strategy is correctly identifying the transactions that might qualify for BAPCPA provisions. Specifically, both 503(b)(9) and 546(c) require that the seller provide goods in the ordinary course of business; thus, my treatment group represents transactions for the exchange of goods. Because BAPCPA does not provide a definition of "goods," courts have generally looked to the definition of goods in Section 2-105(1) of the UCC. While there is uniform agreement that vendors of services are treated as general unsecured claimants, lack of clarity in the UCC definition of goods has resulted in disagreement on whether a non-service item qualifies for "goods" under BAPCPA. Therefore, judgment must be used to identify the treatment and control samples.

Importantly, in the case that vendors sell a combination of goods and services, those claims are bifurcated where possible. This highlights the importance of assigning treatment at the contract or transaction level rather than at the

22 Most contracts do not include information on credit limits; thus, the contracts only allow me to investigate variation in the duration of credit rather than the volume of credit offered.

23 Credit2B strictly enforces a rule that all member firms must upload receivables data at a minimum monthly frequency. If suppliers violate this rule, they lose access to all credit information provided by Credit2B. In discussions with management, they also require their members to upload data for all existing customers. This mandate is more difficult to enforce since not all customer relationships are observable.

24 Discussions with the Office of Corporate Finance at the SEC suggest that the materiality threshold is around 10% of sales.

25 Note that the smaller number of suppliers relative to customers is driven by the structure of the Credit2B business model, which provides information about customers to member firms (suppliers). All member firms report their transactions with customers, but they do not report their transactions with their own suppliers. The sample is highly skewed, with a few very large suppliers causing the averages to be high.

26 Credit2B shared data with Costello on condition of anonymity of the underlying companies. Therefore, firm names, locations, and sample minimums and maximums are not reported per the agreement. Further discussion of the Credit2B data is provided in the Online Appendix.

Table 1
Sample summary statistics versus Compustat universe.
This table presents summary statistics for the samples used in the analyses in the paper. Panel A presents the summary statistics for the sample of contracts, and Panel B presents the summary statistics for the sample of trade transactions. *Size* is measured as the log of total assets, *Cash* is measured as cash and cash equivalents scaled by total assets, *Debt* is measured as the long-term debt scaled by assets, and *Age* is measured as the number of years the firm has appeared in the Compustat file. In addition to these control variables, I include variables that are related to trade credit. *Payment days* are gathered from the contracts and represent the number of days past the invoice that the trade credit is due, and *Current receivable* is the volume of trade credit offered to a customer on each transaction, gathered from the Credit2B sample. *Unique suppliers* and *Unique customers* are counts of the unique suppliers and customers in each sample. The last column in both panels provides sample means for the population of Compustat firms.

(a) Panel A: Summary statistics for contracts sample

	N	*Mean*	*Q1*	*Median*	*Q3*	*Compustat mean*
Payment days	515	28.685	22.5	27	33.6	
Unique suppliers	169					
Unique customers	312					
Size	432	6.719	4.304	7.116	9.191	7.865
Cash	432	0.150	0.027	0.078	0.170	0.091
Debt	432	0.327	0.063	0.198	0.340	0.284
Age	432	19.319	7	11	24	27.429

(b) Panel B: Summary statistics for trade sample

	N	*Mean*	*Q1*	*Median*	*Q3*	*Compustat mean*
Current receivable	6,433,494	31,412.8	708	2800	12,687	
Unique suppliers	1000					
Unique customers	993,615					
Size	514,059	7.498	6.519	7.753	8.513	7.212
Cash	514,059	0.080	0.041	0.075	0.115	0.145
Debt	514,059	0.310	0.168	0.278	0.354	0.301
Age	514,059	31.14	17	30	46	26.175

industry level. Treatment is assigned based on product descriptions. In the contracts sample, items are described in a “Specifications” section of the contract. In the trade sample, items are described for each transaction in a field titled “description,” which is provided by the supplier. I use textual analysis to extract keywords in the product descriptions and then rely on the assessments of two separate corporate bankruptcy attorneys to assign treatment. Details of the classification are outlined in the Online Appendix and are based on the UCC definition of goods and the attorney’s knowledge of case law that sets precedent.[27]

Table 2 presents descriptive statistics comparing treated and control firms in the pre-period. My assumption is that the treatment and control groups are similar along most dimensions, except for the liquidation value of their trade collateral. Panel A reports the summary statistics for the contracts sample, and Panel B reports the summary statistics for the control group. For both the contracts and the trade samples, the treatment and control groups are very similar. There are no significant differences in the size, cash, and age of the firms. The treatment firms in the trade sample are more highly levered, which might be expected given that the treatment firms produce durable goods, while the control firms do not. Further, the treatment firms extend more credit relative to control firms, which again might be an artifact of the types of products offered. Overall, the results in Table 2 suggest that the treatment and control firms are similar along most dimensions. It’s important to note, however, that the lack of clarity in the definition of “goods” may induce noise in the assignment to treatment and may not accurately reflect ex post court rulings. However, I have no reason to believe that the classification scheme would bias the results. Further, I control for a vector of firm characteristics interacted with the *Post* indicator, and I include fixed effects to capture differences in the treatment and control groups. Thus, I have taken measures to ensure that treatment is assigned without bias and that differences in the treatment and control groups do not drive the results.

6. Results

6.1. The effect of BAPCPA on trade credit

First, I test the hypothesis that when suppliers place a higher liquidation value on the goods sold, they lend more to their customers. Before I move to the empirical specification, I plot the impact of BAPCPA on contractual trade days in Fig. 3. The plot shows that treatment and control firms follow a similar trend in trade credit duration during the pre-period but then diverge around BAPCPA. The new bankruptcy code impacted cases filed beginning October 2005. It is possible, however, that parties to the contractual agreements anticipated the enactment of BAPCPA

[27] Keywords are extracted from each transaction in the contracts sample and in the trade sample. The resulting classification scheme results in the following keywords that are classified in the control sample (i.e., they do not qualify for 503(b)(9) or 546(c) treatment): service, consult, electricity, rent, insurance, distribution, printing, trucking, design, power, credit, petroleum, loan, engineering, welding, horticulture, research, water, marketing, advertising, hauling, packaging, ship, inspect, and tax. Any transaction including a description that uses one of these words is classified as the control group, and all other transactions are classified as the treatment group.

Table 2
Covariate balance in the preperiod.
This table reports the descriptive statistics for both the treated and the control firms in the preperiod. Panel A presents the covariate balance for the sample of contracts, and Panel B presents the covariate balance for the sample of trade transactions. The variables include all control variables used in the regressions as well as the main dependent variables of interest. Specifically, the *Number of customers per supplier* is a count of the number of unique customers that the supplier sells to in a given year (contracts sample) or in a given month (trade sample). The *Equal weighted customer credit rating/Score* is an equally weighted credit rating or credit score of the supplier's customer portfolio in a given year (contracts sample) or in a given month (trade sample). The *Value weighted customer credit score* is a summary credit score of the supplier's customer portfolio in a given month, weighted by the share of credit sales to each customer that month. The *Customer concentration (HHI)* is a measure of lending concentration for a supplier in a given month. Finally, *Credit check* is a count of the number of times that the supplier checks the credit score of their customers in a given month. The Difference column calculates the difference in means for the treatment and the control group. ***, **, and * represent significant differences in the treatment and control groups at the 1%, 5%, and 10% level, respectively.

(a) Panel A: Covariate balance for contracts sample

	Treated	*Control*	*Difference*
Payment days	26.593	22.307	4.286*
Number of customers per supplier	1.638	1.813	−0.175
Equal weighted customer credit rating	9.415	9.510	−0.095
Size	6.484	6.104	0.380
Cash	0.161	0.160	0.001
Debt	0.289	0.253	0.036
Age	20.056	18.655	1.401

(b) Panel B: Covariate balance for trade sample

	Treated	*Control*	*Difference*
Current receivable	11,728	5963	5765***
Number of customers per supplier	12,610.65	13,852.70	−1,242.05*
Equal weighted customer credit score	79.732	80.868	−1.136
Value weighted customer credit score	61.461	63.397	−1.936
Customer concentration (HHI)	0.010	0.012	−0.002
Credit check	6.174	5.892	0.282
Size	7.366	7.066	0.300
Cash	0.079	0.081	−0.002
Debt	0.345	0.130	0.215***
Age	32.071	30.288	1.783

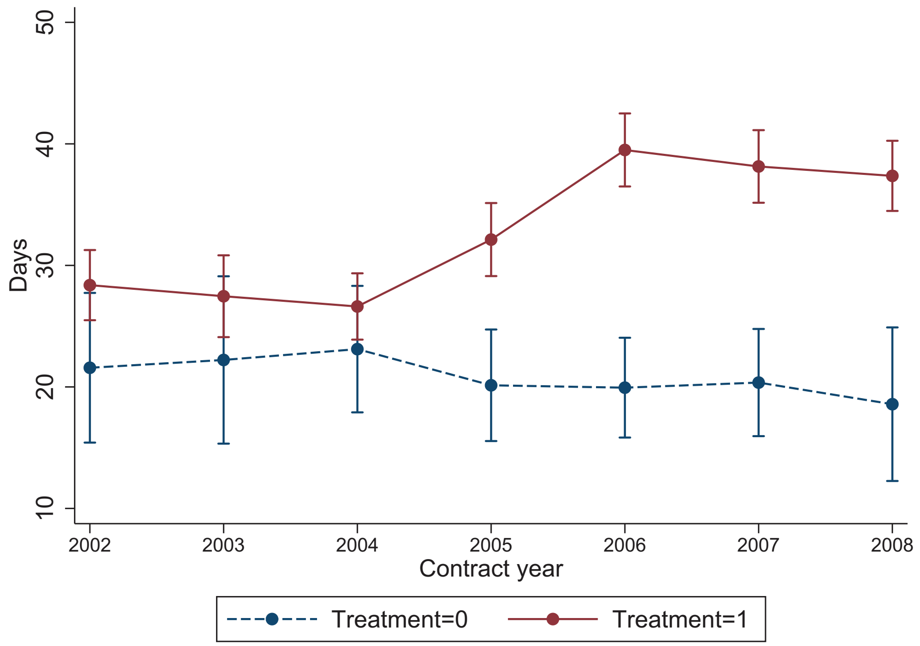

Fig. 3. The impact of BAPCPA on contractual trade credit days.
This figure plots the number of days of credit after invoice that the receivable is due (*Payment days*) over time. *Payment days* are collected from the contractual agreements between the supplier and the customer. The treatment sample (*Treatment*=1) is represented by the solid line and includes all goods suppliers, as defined in Section 5.4 and in the Online Appendix. The control sample (*Treatment*=0) is represented by the dashed line and includes all suppliers that do not qualify for 503(b)(9) and 546(c) treatment under BAPCPA.

Table 3
The impact of BAPCPA on trade credit duration.
This table reports the regression results from Eq. (1) where *Payment days* is the number of days of credit after invoice that the receivable is due. *Payment days* are collected from the contractual agreements between the supplier and the customer. *Treatment* includes the sample of goods suppliers that qualify for 503(b)(9) and 546(c) treatment under BAPCPA, as defined in Section 5.4 and in the Online Appendix, and *Post* is an indicator variable equal to one for all contracts that commence after the enactment of BAPCPA. The *Pre* variable picks up the observations in the period prior to treatment. All other control variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Payment days			
	(1)	(2)	(3)	(4)
Treatment*Post	7.248**	9.878***	10.780**	9.381*
	(3.426)	(3.702)	(4.364)	(5.524)
Treatment*Pre		14.560	9.553	12.690
		(10.285)	(6.274)	(12.280)
Size*Post			0.320	−1.395
			(0.814)	(1.289)
Cash*Post			−11.230	−15.980
			(7.770)	(18.630)
Age*Post			−0.038	0.179
			(0.076)	(0.197)
Debt*Post			−0.766	−0.365
			(0.468)	(1.669)
Observations	515	515	432	251
Seller FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Buyer FE	N	N	N	Y
Adj. R^2	0.368	0.374	0.384	0.580

before that date and formed the contract based on the anticipated new law. Indeed, there is some divergence in 2005 but a larger increase in 2006 and thereafter. There appears to be no change in trade terms for non-goods suppliers, suggesting that the divergence in trade credit days can be attributed to a treatment effect.

To formally test the collateral hypothesis, next I estimate Eq. (1) using ordinary least squares. Recall that 503(b)(9) improved the trade creditor's collateral recovery values, and 546(c) increased the duration of the reclamation reach-back period. Thus, to capture the lending behavior of suppliers, I investigate both the duration of the trade credit, measured as the ex-ante agreed upon number of days after invoicing that the customer has to pay, and the amount of trade credit, measured as the ex post volume of credit extended from the seller to the customer. The primary coefficient of interest is the *Treatment*Post* term, which captures the effect of the difference-in-differences. Standard errors are clustered at the supplier level.

The results from estimating Eq. (1), where the dependent variable is the number of days of credit offered, are reported in Table 3. Column (1) measures the treatment effect without controls. Consistent with predictions from the collateral theory of trade credit, I find that treatment firms increase the number of days of trade credit after BAPCPA by seven more days than control firms. In column (2), I add an indicator for the preperiod interacted with the *Treatment* indicator to assess any pretreatment trends. The results indicate that there are no pretreatment differences in the treatment and control groups. In column 3, I add control variables interacted with the *Post* indicator. The control variables help to alleviate the concern that my results might be driven by pretreatment variation between the treatment and control groups, and they help to prevent bias if the treatment and control groups respond differently to macroeconomic fluctuations. The controls do not enter into the regression separately because they are absorbed by the fixed effects. After controlling for other known factors that influence trade credit, the main result continues to hold. Further, in all specifications, I include supplier fixed effects to control for any time-invariant characteristics of the supplier that may drive trade credit policy. Year fixed effects are included in all specifications to capture any macroeconomic fluctuations that might correlate with lending decisions. In column 4, I add buyer fixed effects to control for time-invariant characteristics of the buyer that might correlate with trade credit taken. Since there are relatively few redundant buyers, the fixed effect reduces my sample size by half, though the main result remains significant. Overall, the results in Table 3 suggest that treated suppliers increase trade credit by 7 to 11 days more than the control group. Relative to the sample average trade days of 29, the magnitude of the effect is economically significant. Since Section 546(c) of BAPCPA materially changed the reach-back period for reclamation from 10 to 45 days, the large magnitude of the results seems reasonable.

Next, I move to how BAPCPA influences the volume of trade credit extended. Again, I start by plotting impact of BAPCPA on the volume of trade credit lent in Fig. 5. While the treatment firms offer more credit than the control firms in the preperiod, they appear to follow parallel trends. In this case, the effect of BAPCPA is sharp and dramatic. Starting at the end of 2005, goods suppliers begin offering more credit on each transaction. No such trend appears for the non-goods suppliers, again suggesting that the divergence in trade credit volume can be attributed to a treatment effect.

The collateral theory predicts that if BAPCPA increased liquidation values, suppliers will be willing to extend more credit. Therefore, in Table 4, I estimate Eq. (1), where the dependent variable is the volume of credit extended. The level of observation is a unique buyer-supplier-transaction, so the dependent variable represents the credit sales extended by a given supplier to a given buyer on a single transaction, as reported in the monthly receivable logs collected by Credit2B. Note that the balance of current receivables represents new credit sales made to a particular customer in a given month. Since the receivables data is highly positively skewed, I estimate the regression using logs.

The first column of Table 4 measures the treatment effect without control variables. Consistent with the collateral theory of trade credit, I find that suppliers impacted by BAPCPA increase the credit offered to customers by 70% more than the control group of suppliers. Moving from columns 1 through 5, the empirical specification becomes more rigorous. In column 2, I add a pretreatment trend, and I add control variables interacted with the *Post* indicator in columns 3–5. While the inclusion of control

Table 4
The impact of BAPCPA on trade credit volume.
This table reports the regression results from Eq. (1) where *Current receivable* is the log of the amount of trade credit offered to a customer on each transaction. *Current receivable* is collected from transaction data provided by Credit2B. *Treatment* includes the sample of goods suppliers that qualify for 503(b)(9) and 546(c) treatment under BAPCPA, as defined in Section 5.4 and in the Online Appendix, and *Post* is an indicator variable equal to one for all contracts that commence after the enactment of BAPCPA. The *Pre* variable picks up the observations in the period prior to treatment. All other control variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Current receivable				
	(1)	(2)	(3)	(4)	(5)
Treatment*Post	0.697***	0.603***	0.417**	0.385**	0.309***
	(0.084)	(0.168)	(0.177)	(0.189)	(0.147)
Treatment*Pre		−0.166	−0.663	−0.667	−0.714
		(0.153)	(0.563)	(0.638)	(0.677)
Size*Post			0.036	0.030	0.078**
			(0.026)	(0.029)	(0.036)
Cash*Post			−1.325	−1.590*	−0.953
			(0.743)	(0.832)	(1.359)
Age*Post			−0.004*	−0.005**	−0.002
			(0.002)	(0.002)	(0.003)
Debt*Post			0.029	0.032	0.049
			(0.061)	(0.064)	(0.170)
Observations	6,433,494	6,433,494	514,059	508,878	72,757
Seller FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Buyer FE	Y	Y	Y	Y	Y
BuyerXSeller FE	N	N	N	Y	Y
BuyerXMonth FE	N	N	N	N	Y
Adj. R^2	0.712	0.712	0.713	0.703	0.752

variables significantly impacts my sample size, the main effect continues to hold.

Due to the volume of data offered in the Credit2B sample, I am able to include additional fixed effects to better isolate the causal channel linking the change in liquidation values to the trade credit offered. In all specifications in Table 4, I include seller, buyer, and month fixed effects. The buyer and seller fixed effects absorb the time-invariant characteristics of the contracting parties that may impact trade credit, while the month fixed effect should control for macroeconomic characteristics that might impact trade. In column 4, I add a buyerXseller fixed effect to isolate the within-relationship change in trade credit offered after BAPCPA. The results, though slightly economically smaller, are still significant; treated suppliers increase the volume of credit 38% more than the control firms. Finally, in column 5, I add a buyerXmonth fixed effect to absorb any time-varying customer characteristics like product or credit demand. In all specifications, I find a positive and significant coefficient on the *Treatment*Post* term, suggesting that suppliers experiencing an increase in the liquidation value of their collateral due to BAPCPA are willing to extend more credit, relative to the control sample of suppliers.

6.2. The effect of an improvement in reclamation rights as distinct from priority rights

As discussed in Section 3.3, 546(c) represents an improvement in collateral values through a reduction in the costs of reclaiming goods, whereas 503(b)(9) represents a joint improvement in priority rights and collateral values through the allowance of an administrative claim for the value of the goods. However, since both 503(b)(9) and 546(c) impact the same group of goods suppliers, the treatment effect captures both the perceived benefits from reclamation rights alone and from the joint improvement in priority and collateral. One may be concerned that the results are not operating solely through the collateral channel; for example, if suppliers only value the joint benefit of priority and collateral liquidation, then the results of the paper should be interpreted more generally through the lens of an improvement in the protection of trade creditors.

To capture whether suppliers value collateral reclamation protection alone, I take advantage of cross-sectional variation in the supplier's expected ability to reclaim goods. In particular, a vendor will lose his right to reclaim any goods that the customer sells before receiving the vendor's reclamation demand. I match each customer in the sample to Compustat by name and location, where possible, and I calculate the average days that their goods sat in inventory over the preperiod.[28] Since 546(c) allowed reclamation of goods for up to 45 days after sale, I assign an indicator variable (*Seizable*) equal to one if the supplier's transaction is with a customer that has an inventory turnover ratio of greater than 45 days, zero otherwise.

I estimate Eq. (1), where the dependent variable is either the number of days of credit offered or the log of the current receivable offered. I capture cross-sectional variation in the treatment effect by augmenting the

[28] Days in inventory are calculated by dividing 365 by the inventory turnover ratio. The inventory turnover ratio is calculated as cost of goods sold divided by ending inventory. I calculate days in inventory on an annual basis for each customer in the sample and then take the average days in inventory for each customer over the preperiod.

Table 5
BAPCPA on trade duration and volume: cross-section.
This table reports the regression results from Eq. (1) where *Payment days* is the number of days of credit after invoice that the receivable is due, and *Current receivable* is the log of the amount of trade credit offered to a customer on each transaction. *Treatment* includes the sample of goods suppliers that qualify for 503(b)(9) and 546(c) treatment under BAPCPA, as defined in Section 5.4 and in the Online Appendix, and *Post* is an indicator variable equal to one for all contracts that commence after the enactment of BAPCPA. *Seizable* is an indicator variable equal to one if the supplier is transacting with a customer whose average days in inventory over the preperiod are greater than 45 days. The *Pre* variable picks up the observations in the period prior to treatment. All other control variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Payment days	*Current receivable*
	(1)	(2)
Treatment*Post	10.787	0.795*
	(12.506)	(0.413)
Treatment*Post*Seizable	15.581**	0.666***
	(4.921)	(0.157)
Treatment*Pre	16.722	0.197
	(13.743)	(0.237)
Size*Post	−3.781*	−0.004
	(2.094)	(0.054)
Cash*Post	−32.978	1.795
	(35.785)	(1.260)
Age*Post	0.295**	0.005
	(0.110)	(0.004)
Debt*Post	−18.281	0.918
	(12.772)	(0.531)
Observations	136	2,634
Seller FE	Y	Y
Month FE	Y	Y
Buyer FE	Y	Y
BuyerXSeller FE	N	Y
BuyerXMonth FE	N	N
Adj. R^2	0.801	0.813

specification with the *Seizable* indicator. If suppliers value reclamation rights alone, then I expect a positive coefficient on the triple interaction term, *Treatment*Post*Seizable*. Results are reported in Table 5 and indicate that treated suppliers extend more credit (both in duration and amount) to their customers that they expect to have more seizable collateral. This is consistent with the idea that reclamation rights, by themselves, offer incremental value to suppliers relative to 503(b)(9) that bundles both priority and collateral protections together, though both provisions appear to influence suppliers to some degree. However, results from these tests should be interpreted with caution since the sample is much smaller than the main analyses.[29]

6.3. The effect of BAPCPA on the portfolio of borrowers

While the main results show that suppliers impacted by an increase in collateral values increase the volume and duration of trade credit offered, the question remains: to whom? Prior work suggests that changes in trade credit terms can have important economic consequences like changes in sourcing decisions (Breza and Liberman, 2017), market entry (Barrot, 2016), and economic spillovers (Jacobson and Schedvin, 2015; Costello, 2018). However, the prior literature on trade credit generally ignores the question of who suppliers lend to, likely due to the inability to observe trade counterparties.

In the BAPCPA setting, it is important to understand who suppliers lend to. One possibility is that suppliers increase credit equally to customers in the current lending portfolio. However, the prior literature on securitization suggests that it may incentivize lenders to lend to riskier customers since the suppliers don't bear the full loss in the event of default. For example, Mian and Sufi (2009) find that mortgage securitization allowed banks to lend to riskier homeowners.

The results in the main specification suggest that at least a portion of the increase in lending was passed to suppliers' existing customers. For example, column 4 of Table 4 shows that within a buyer-seller pair, treated suppliers increase credit more to their existing customers. To shed further light on this, Fig. 4 plots the change in duration of trade credit along the intensive and extensive margins. For visual clarity, I only plot the margins for *Treatment* suppliers. The trend lines show that around BAPCPA, treated suppliers increase the duration of trade credit to both existing and new customers. Fig. 6 plots the change in the volume of credit that treated suppliers offer, separated into the intensive and extensive margins. The figure shows that treated suppliers increase the volume of trade credit offered around BAPCPA; the increase appears more significant within the sample of existing customers, but there is a notable increase in the volume of credit offered to new customers as well.

To formalize, I test whether treated suppliers change their customer portfolio after BAPCPA by estimating Eq. (1), where the dependent variable captures customer concentration. In the contracts sample, I use a simple count of the number of customers that a given supplier has at one time as a measure of customer concentration. In the trade sample, in addition to customer count, I calculate customer concentration using the Herfindahl-Hirschman index, where customer market share is calculated as the percentage of a supplier's total credit sales made to a given customer each month.[30]

The results of the regression are reported in Table 6. Since the number of customers per supplier-month is highly skewed, the dependent variables in columns 1 and 2 are calculated as the log of the number of customers. Consistent with higher collateral protections increasing suppliers' incentives to lend to a more diverse pool of borrowers, the results show that treated suppliers increase the number of customers that they lend to more than control firms. The number of contractual customers that BAPCPA suppliers lend to increases by 27% more than control sup-

[29] The sample is constrained by the number of customers matched to Compustat, data availability for these customers over the preperiod, and too few observations to include buyer-seller fixed effects in column (1) and buyer-month fixed effects in column (2).

[30] Note that since I have the volume of credit sales in the trade sample, I am able to calculate a concentration ratio. Due to the absence of volume data in the contracts sample, I am unable to calculate a concentration index.

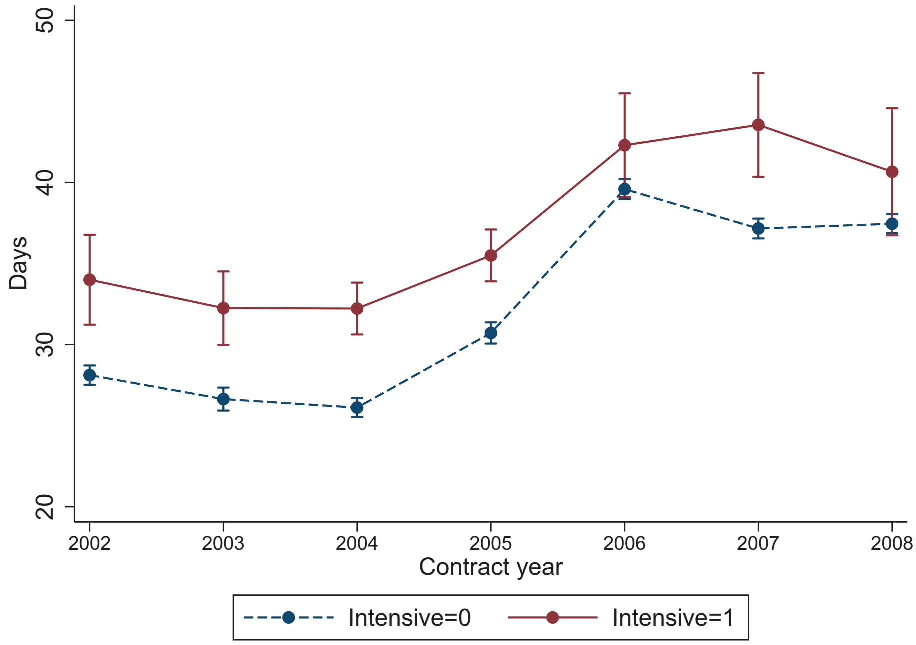

Fig. 4. Change in contractual trade credit days around BAPCPA: intensive and extensive margins.
This figure plots the number of days of credit after invoice that the receivable is due (*Payment days*) over time for the subsample of suppliers in the treatment group. The treated suppliers include the sample of firms that qualify for 503(b)(9) and 546(c) treatment under BAPCPA, as defined in Section 5.4 and in the Online Appendix. The sample of treated suppliers is split into the intensive margin (*Intensive=1*), which is represented by the solid line and the extensive margin (*Intensive=0*), which is represented by the dashed line. The intensive margin requires that the supplier-customer relationship persists for the sample period, while the extensive margin allows new customers to enter at any stage.

pliers, and the number of trade customers that BAPCPA suppliers lend to increases by 18% more than control suppliers. Finally, column 3 reports the results where the dependent variable is the Herfindahl index of customer concentration. After BAPCPA, treated suppliers show a decline in monthly customer concentration relative to control suppliers.[31]

6.3.1. The effect of BAPCPA on customer portfolio risk

To further test whether increased securitization enhances the supplier's incentive to take on risk, I investigate changes in the riskiness of suppliers' lending portfolios. Evidence from the 2007-2008 financial crisis suggests that lenders, through the securitization of mortgage backed securities, lent more money to borrowers of poor credit quality. Since loan origination risk was not fully borne by the originator, banks had incentives to reduce lending standards below where they otherwise would. For example, several empirical studies show that securitization of MBS led to lower lending standards (Keys et al., 2010; Mian and Sufi, 2009; Nadauld and Sherlund, 2013). Alternatively, it is possible that BAPCPA had no impact of the incentive to lower lending standards. Unlike mortgage backed securities, which were typically fully offloaded from the originator's balance sheet, trade credit risk is often retained by the supplier, and BAPCPA did not promise 100% recovery. Following similar arguments, Benmelech et al. (2012) find that securitization in the form of collateralized loan obligations did not materially change the risk portfolio of borrowers receiving credit.

To test the impact of increased trade credit collateral value on lending standards, I estimate Eq. (1), where the dependent variable captures the credit quality of the pool of customers that suppliers offer trade credit to. Table 7 reports the results. In column 1, the dependent variable is the average (equal-weighted) credit rating of each supplier's customer in a given year. I merge the customer's GVKEY to Mergent FISD to obtain the credit rating in the year that the contract started. Letter ratings are converted to a numeric scale ranging from 1 to 22, where 1 represents the highest credit rating and 22 represents the lowest. The results in column 1 show that the credit risk of treated suppliers' portfolio goes up by five points more than the control sample after BAPCPA, indicating that the additional collateral protection offered by BAPCPA incentivizes treated suppliers to take on additional risk.

Columns 2 and 3 investigate the risk of a suppliers' portfolio in the trade sample. Since Credit2B is in the business of credit scoring, I use the individual customer's Credit2B credit score as the dependent variable. Credit scores range from 0 to 100, where higher scores indicate more creditworthy customers. The scores are based on a wide range of data including payment histories, financial data, and other news sources. In column 2, I use an equally

[31] Note that I do not include customer fixed effects in these regressions since I am not interested in within-customer effects but rather the supplier's incentives to grow their customer portfolio.

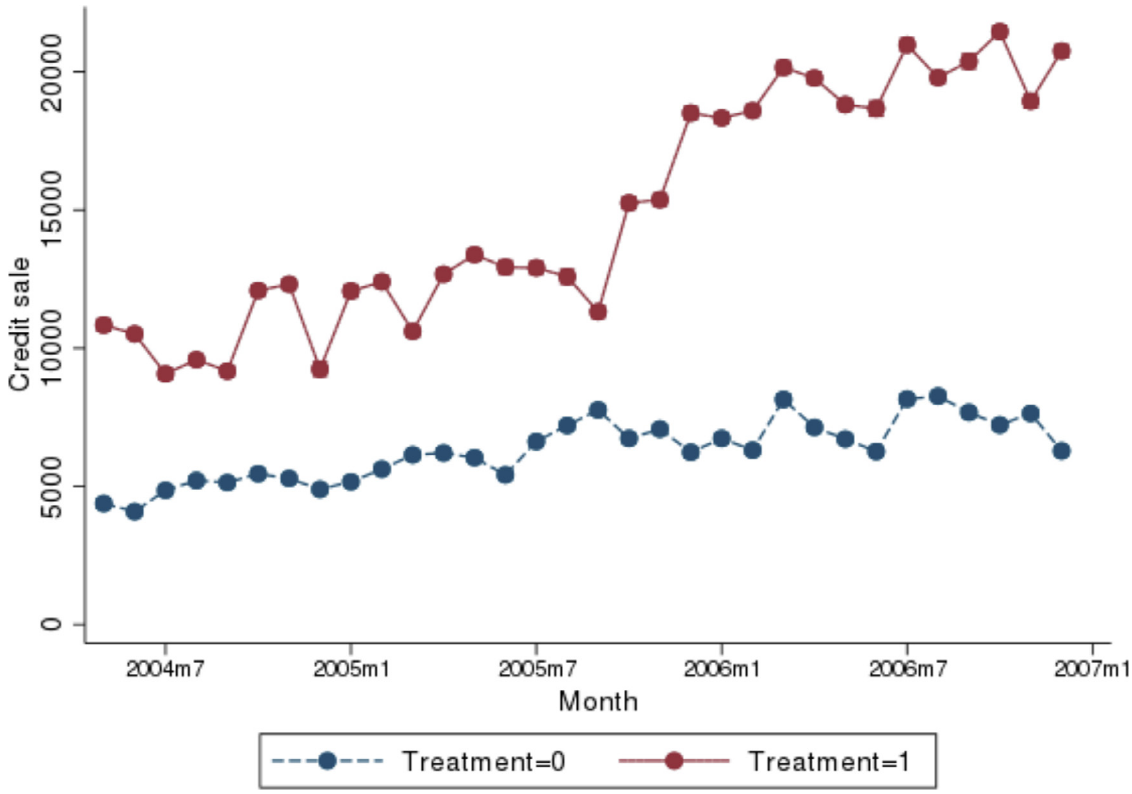

Fig. 5. The impact of BAPCPA on the amount of trade credit extended.
This figure plots the amount of trade credit offered to a customer (*Current receivable*) over time. *Current receivable* is the log of the amount of trade credit offered to a customer on each transaction. The treatment sample (*Treatment*=1) is represented by the solid line and includes all goods suppliers, as defined in Section 5.4 and in the Online Appendix. The control sample (*Treatment*=0) is represented by the dashed line and includes all suppliers that do not qualify for 503(b)(9) and 546(c) treatment under BAPCPA.

Table 6
The impact of BAPCPA on customer concentration.
This table reports the regression results from Eq. (1) where the dependent variable is a measure of customer concentration. In column 1, *Number contractual customers* is the log of the number of customers that a supplier has an outstanding contract with in each year ("contracts data"). In column 2, *Number trade customers* is the log of the number of customers that a supplier transacts with in each month ("trade data"). Finally, in column 3, *Customer concentration* is measured as the HHI of credit sales made by suppliers to their customers in a given month ("trade data"). All other variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Number contractual customers	*Number trade customers*	*Customer concentration*
	(1)	(2)	(3)
Treatment*Post	0.268**	0.180*	−0.019**
	(0.133)	(0.099)	(0.008)
Size*Post	−0.056	−0.075	−0.002
	(0.040)	(0.090)	(0.002)
Cash*Post	−0.499	2.520	−0.106**
	(0.740)	(1.529)	(0.038)
Age*Post	−0.003	0.007	−0.000*
	(0.006)	(0.006)	(0.000)
Debt*Post	−0.047	−0.382*	−0.004
	(0.081)	(0.151)	(0.003)
Observations	432	952,296	952,296
Seller FE	Y	Y	Y
Year/Month FE	Y	Y	Y
Buyer FE	N	N	N
Adj. R^2	0.605	0.974	0.748

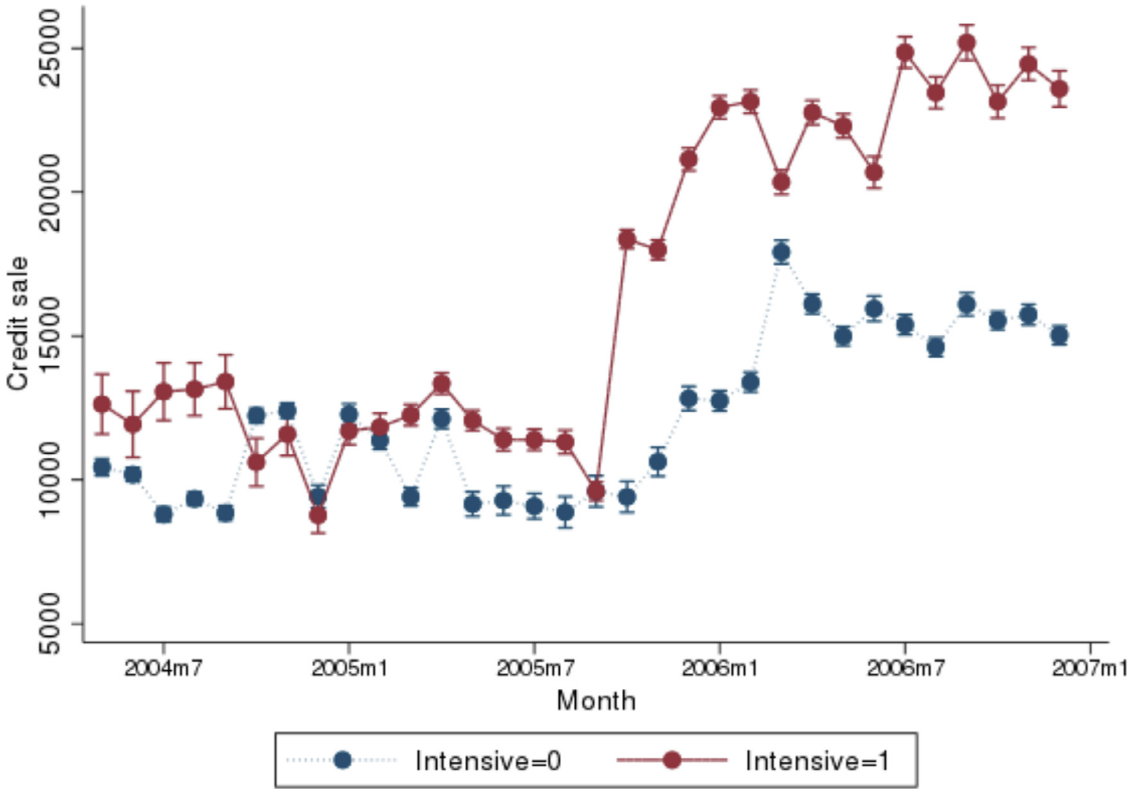

Fig. 6. Change in trade credit around BAPCPA: intensive and extensive margins.
This figure plots the amount of trade credit offered to a customer (*Current receivable*) over time for the subsample of suppliers in the treatment group. The treated suppliers include the sample of firms that qualify for 503(b)(9) and 546(c) treatment under BAPCPA, as defined in Section 5.4 and in the Online Appendix. The sample of treated suppliers is split into the intensive margin (*Intensive=1*), which is represented by the solid line and the extensive margin (*Intensive=0*), which is represented by the dashed line. The intensive margin requires that the supplier-customer relationship persists for the sample period, while the extensive margin allows new customers to enter at any stage.

Table 7
The impact of BAPCPA on the credit quality of the customer portfolio.
This table reports the regression results from Eq. (1) where the dependent variable is a measure of customer credit quality. In column 1, *Credit rating* is an equally weighted credit rating of the supplier's customer portfolio in a given year ("contracts data"). Credit ratings are obtained by matching the customer sample to Mergent FISD, and numerical ratings are assigned on a scale from 1 to 22, where 1 represents the highest credit rating. In column 2, *Credit score* is an equally weighted credit score of the supplier's customer portfolio in a given month ("trade data"). Credit scores are based on the proprietary scoring algorithm by Credit2B. They range from 1 to 100, where higher scores indicate more creditworthy customers. Finally, in column 3, *Value weighted customer credit score* is the credit score of the supplier's customer portfolio in a given month, weighted by the share of credit sales to each customer that month ("trade data"). All other variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Credit rating	*Credit score*	*Weighted average credit score*
	(1)	(2)	(3)
Treatment*Post	5.521**	−11.790***	−4.565***
	(2.288)	(1.888)	(0.367)
Size*Post	0.573	0.705**	−0.232***
	(0.860)	(0.263)	(0.041)
Cash*Post	1.761	−110.400***	11.850***
	(9.410)	(24.040)	(2.030)
Age*Post	−0.134**	−0.293***	0.0147**
	(0.048)	(0.063)	(0.007)
Debt*Post	9.494**	2.324	0.109
	(2.985)	(1.760)	(0.186)
Observations	152	17,508	17,508
Seller FE	Y	Y	Y
Year/Month FE	Y	Y	Y
Adj. R^2	0.743	0.112	0.886

weighted customer portfolio as in column 1. In column 3, I calculate a value-weighted customer risk measure based on the proportion of the total trade credit that is offered to a given customer in each month. The results from both specifications confirm that the increase in collateral values following BAPCPA decreased suppliers' lending standards. For example, column 2 suggests that the credit quality of treated suppliers' portfolios declined by 11 more points, relative to control suppliers. Relative to the pretreatment average credit score of 80, the magnitude is economically significant.[32]

6.4. The effect of BAPCPA on incentives to monitor

BAPCPA improved the trade creditor's rights to collateral and increased the liquidation value of collateral. The analysis in Section 6.1 confirms that this enhanced the trade creditor's willingness to lend. Next, I analyze how the shift in collateral value influenced the trade creditor's incentives to monitor.

The prior literature suggests that the relation between collateral and monitoring could go either way. Manove et al. (2001) model the link between creditor rights and monitoring incentives and find that collateral and screening are substitutes. The protection offered from collateral reduces the risk of the loan and therefore decreases the investor's incentive to monitor. Empirical literature studying the financial crisis of 2007-2008 generally shows that securitization led to lax screening and monitoring (Keys, Mukherjee, Seru and Vig, 2010; Keys, Piskorski, Seru and Vig, 2012; Nadauld and Sherlund, 2013). On the other hand, Rajan and Winton (1995) model how collateral might improve lenders' incentives to monitor. When the lender's payoff is sensitive to the borrower's financial health, collateral is only effective if the lender can monitor its value. Cerqueiro et al. (2016) show that for a sample of Swedish banks, lenders responded to a reduction in collateral values by reducing their monitoring efforts, suggesting a complementary relation between collateral and monitoring. Further, Section 6.3.1 shows that the law incentivized trade creditors to lend to riskier clients. To the extent that BAPCPA increased portfolio credit risk, we might observe trade lenders increase the monitoring of all customers.

To provide evidence on the relation between collateral and monitoring, I investigate the impact of BAPCPA on trade creditor's monitoring frequency. It is inherently difficult to capture monitoring effort since it is mostly unobservable. However, for the trade sample of Credit2B firms, I observe the frequency with which suppliers sign into the system to check the credit scores of their customers. While suppliers may also monitor their customers in other ways, checking credit scores through a third party credit agency is a primary means for monitoring customer performance. To test whether suppliers in the treatment sample change their monitoring behavior after BAPCPA, I estimate Eq. (1), where the dependent variable captures monitoring intensity. Specifically, *Credit check* is a count variable of the number of times in a given month that a supplier checks credit scores. Table 8 reports the results. Consistent with the hypothesis that collateral and monitoring are substitutes, I find that treated suppliers decrease monitoring by two to four fewer credit checks than control suppliers after BAPCPA. When compared to the sample mean of six credit checks per month, this result is economically significant. In moving from column 1 through column 4, I add additional fixed effects to the regression, but the main variable of interest remains statistically and economically significant. For example, in the last column I add a BuyerXMonth fixed effect, which should control for time-varying characteristics of the buyer such as credit quality and credit demand. The negative relation between collateral and monitoring remains significant, suggesting that the results are attributable to changes in collateral values and are not merely an artifact of changes in time-varying customer characteristics.

Table 8
The impact of BAPCPA on monitoring incentives.
This table reports the regression results from Eq. (1) where *Credit check* is a count of the number of times that the supplier checks the credit score of their customers in a given month. All other control variables are defined in prior tables. Standard errors are clustered at the supplier level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Credit check			
	(1)	(2)	(3)	(4)
Treatment*Post	−3.354**	−2.820**	−2.873**	−1.950*
	(1.688)	(1.386)	(1.319)	(1.393)
Size*Post	−0.285	−0.229	−0.243	−0.308
	(0.203)	(0.218)	(0.251)	(0.221)
Cash*Post	−61.67**	−45.38**	−43.90**	−30.76**
	(22.65)	(19.35)	(20.25)	(14.58)
Age*Post	−0.107*	−0.0765	−0.0732	−0.0377
	(0.0584)	(0.0473)	(0.0512)	(0.0312)
Debt*Post	−8.199***	−7.063***	−6.953***	−5.200**
	(1.344)	(1.457)	(1.518)	(2.037)
Observations	952,296	906,510	902,126	95,818
Seller FE	Y	Y	Y	Y
Buyer FE	Y	Y	Y	Y
Month FE	N	Y	Y	Y
BuyerXSeller FE	N	N	Y	Y
BuyerXMonth FE	N	N	N	Y
Adj. R^2	0.680	0.689	0.624	0.791

7. Additional analyses and robustness

7.1. The BAPCPA and other lenders

Because the objective of the paper is to identify whether an improvement in collateral rights influence the supply of credit from suppliers, thus far I have neglected how the shift in trade creditor rights impact other lenders. However, higher collateral protection for trade creditors might indicate that some other creditors now have less goods to liquidate. Facing lower liquidation values should

[32] Note that I do not include customer fixed effects in these regressions since I am not interested in within-customer effects but rather the supplier's incentives to lend to new, riskier customers. In an untabulated analysis, I find that on the intensive margin, the change in treated firms' portfolio credit risk is not statistically distinguishable from that of the control sample. This suggests that the results are primarily attributable to suppliers taking on new, riskier clients.

result in an ex-ante decline in lending from these creditors if they value collateral protection.

To investigate the impact of BAPCPA on other forms of credit, I match the customers in the sample to S&P Capital IQ by name and location, where possible. The Capital IQ Debt Capital Structure database collects attributes of each type of debt from SEC filings, press releases, company websites, and other sources.[33] Capital IQ provides the amount of each type of debt for firms on an annual and interim basis. Other features of the debt, such as interest rates, are not available or too sparsely populated to be included in the analyses.

Eq. (1) is estimated at the customer/debtor-year level to identify whether other lenders, aside from goods suppliers, adjust their lending behavior to debtors in response to BAPCPA. Customers are assigned to the *Treatment* sample if their trade transactions include the purchase of goods and therefore have suppliers that face higher collateral protection after BAPCPA (i.e., whether they have a transaction linked to a treated supplier). For each customer in my sample, I use Capital IQ to identify their borrowing from senior lenders; junior or subordinated lenders; junior, secured lenders; and junior, unsecured lenders. Each dependent variable is scaled by the customer's total assets to account for variation in the scale of the firm. Since some debtors report their capital structure on an interim basis and some on an annual basis, I take the latest observation for each customer-year. The coefficient of interest is β_2, which should capture the incremental change in the lending behavior of other creditors to the *Treatment* group of debtors after BAPCPA, relative to the change in their lending to the control group.

Table 9 reports the results. There is no significant difference in the change in lending from senior lenders or from junior, secured lenders after BAPCPA to the customers in the treatment group, relative to the control group. However, column (2) suggests that subordinated lenders reduce their credit to treated customers more after BAPCPA, relative to their lending to the control group. When I split the subordinated loans into those that are secured versus unsecured, the effect is primarily coming through debt that is both junior in priority and unsecured by collateral. Therefore, the evidence in Table 9 is consistent with the hypothesis that BAPCPA shifted collateral protection away from lower classes of lenders to trade creditors that sell goods.

7.2. Contemporaneous legislation

The results of the paper are consistent with the story that improvements in repossession rights and liquidation rights encourage trade creditors to lend. However, one concern may be that the increase in trade credit to the treatment group operates either directly or indirectly through contemporaneous legislation that differentially impacts goods suppliers. Contemporaneous legislation could threaten my inferences if it directly targets sellers of goods through another channel, for example, legislation that more generally impacts higher priority lenders. Further, contemporaneous legislation could indirectly impact goods suppliers if, for example, other forms of credit become less attractive and therefore trade credit demand from goods suppliers goes up.

The other BAPCPA provisions that apply to corporations are discussed in the Online Appendix. Those provisions include the following: (1) Section 366 requires the debtor to provide adequate assurance of future payment to utility providers; (2) Section 1121(d)(2) reduces the exclusivity period that a debtor has to file a plan of reorganization; (3) Section 365(d)(4) limits the time the debtor has to assume or reject a lease; (4) Section 503(c)(1) prohibits administrative claims for payments to insiders; and (5) Section 1102(b)(3) requires creditors' committees to provide access to information to creditors who are not on the committee. As discussed in the Appendix, none of these other provisions directly mention goods suppliers as their target audience.

One possible concern is that other legislation impacts goods suppliers but does so through a channel that is distinct from the collateral channel. For example, any legislation that improves the rights of more senior lenders would also differentially improve the rights of goods suppliers (who now hold a 503(b)(9) claim) relative to non-goods suppliers. By shortening the exclusivity period of the debtor, Section 1121(d)(2) potentially offers more senior lenders a differential benefit from early intervention in the reorganization process. If the trade credit results that I show are operating through the direct impact of contemporaneous legislation impacting senior lenders, I would expect an increase in lending from all senior lenders to the treatment group rather than simply from goods suppliers. The results shown in Table 9 suggest otherwise; column (1) shows that there is no difference in lending to the treatment group from senior lenders, which represents primarily senior bank debt, relative to their lending to the control group. This evidence helps to mitigate the concern that the results are operating through a priority channel.

A second concern is that BAPCPA makes another form of credit less attractive, and it thereby improves the debtor's demand for credit from goods suppliers (i.e., the indirect or spillover channel). As discussed in the Online Appendix, to my knowledge there was no legislation that directly reduced the protections of a different group of creditors. Indeed, BAPCPA was viewed as a general improvement in creditor rights.[34] However, I recognize that it is difficult to observe how BAPCPA influenced the supply and demand of all other forms of credit. For example, off-balance sheet debt is typically unobservable to the researcher, and the analyses in Table 9 are limited to a subset of observable firms.

There is some comfort in the fact that it is more difficult to explain the confluence of evidence presented in the paper based on a spillover channel. For example, the

[33] Though there is limited information in Capital IQ, it appears that most of the reports cover only bank debt and not other liabilities such as trade credit or off-balance-sheet debt.

[34] Of course, by improving the rights of some creditors, others may lose out since the size of the pie is fixed. I interpret the evidence in Table 9 as a spillover from 503(b)(9) and 546(c) to junior, unsecured lenders since I'm unaware of any legislation that directly reduced the rights of junior, unsecured lenders.

Table 9
The impact of BAPCPA on other credit.
This table reports the regression results from Eq. (1) for the sample of customers in the data set. The dependent variable is the annual volume of the customer/borrower's outstanding debt obtained from the Capital IQ capital structure database, scaled by their total assets. *Treatment* is set equal to one for the customers in the sample that purchase goods (i.e., are linked to at least one treated supplier), as defined by BAPCPA during the sample period, and zero otherwise. *Post* is set equal to one for periods after October 2005, and zero otherwise. The dependent variable in column (1) is the total senior debt outstanding; the dependent variable in column (2) is the total junior debt outstanding; the dependent variable in column (3) is the total amount of debt outstanding that is both junior in priority and secured by collateral; and the dependent variable in column (4) is the the total amount of debt outstanding that is both junior in priority and is not secured by collateral. Capital IQ defines seniority in a field called LevelTypeId. I classify debt as senior if it has a level of 1, and I classify debt as junior if it has a level of 2-5. Capital IQ defines security in a field called SecuredTypeId. I classify debt as secured if it has an ID of 2 or 4 and unsecured if it has an ID of 3. All other control variables are defined in prior tables. Standard errors are clustered at the customer/borrower level and are reported in parentheses. *** significant at 1%, ** significant at 5%, * significant at 10%.

	Senior (1)	*Junior* (2)	*Junior, Secured* (3)	*Junior, Unsecured* (4)
Treatment*Post	−0.004	−0.053*	−0.001	−0.053**
	(0.005)	(0.027)	(0.012)	(0.024)
Size*Post	−0.001	0.020**	0.007**	0.014**
	(0.001)	(0.007)	(0.003)	(0.006)
Cash*Post	−0.011	0.201	0.022	0.168
	(0.050)	(0.164)	(0.063)	(0.149)
Age*Post	0.000***	−0.001**	0.000	−0.001**
	(0.000)	(0.001)	(0.000)	(0.001)
Debt*Post	−0.064***	−0.518***	−0.203***	−0.315**
	(0.016)	(0.109)	(0.046)	(0.099)
Observations	4920	4920	4920	4920
Buyer FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Adj. R^2	0.605	0.532	0.644	0.497

reduced demand for another form of credit would have to be compensated only by credit from goods suppliers and not from non-goods suppliers or other senior lenders. Further, the spillover would also have to change the behavior of goods suppliers relative to the control group; specifically, it would have to impact their customer portfolio and monitoring activity. Though explaining the full set of results using the spillover channel becomes difficult, inferences should be interpreted with the caution that it is difficult to fully control for potential confounding effects from unintended spillovers from other BAPCPA legislation.

7.3. Is new collateral created?

Though the results in Table 9 are consistent with the idea that BAPCPA resulted in a shift of collateral from junior unsecured lenders toward suppliers providing goods, it is unclear whether this increased the total liquidation value of the customer/debtor or left it unchanged.[35] Longhofer and Santos (2003) and Frank and Maksimovic (1998) argue that because the original seller is better positioned to deal with the collateral than an outside investor, the trade creditor may be willing to lend at a cheaper rate than the bank otherwise would; in other words, suppliers place a higher value on the collateral than other lenders. If this is the case, then allocating collateral to suppliers rather than banks should increase the total liquidation value of the firm. Whether the shift in the law increased total collateral value or simply shifted it among creditors is an empirical question.

Since it's difficult to observe how each of the firm's lenders value the collateral ex-ante, I obtain projected liquidation values of assets from the disclosure statements of firms filing for Chapter 11. Using this data, I test whether BAPCPA resulted in higher firm liquidation values. The empirical method and results are discussed in the Online Appendix, and I fail to show any change in asset liquidation value for debtors filing after BAPCPA for either the treatment or the control group. Therefore, even if suppliers value collateral more than other lenders, the BAPCPA reallocation did not have a quantitatively important impact on firms' total collateral value. The results should be interpreted with caution due to the small sample size, though they seem to imply that BAPCPA simply shifted collateral among creditors.

8. Conclusion and discussion

Much of the prior literature questions the puzzling observation that suppliers often serve as lenders in the presence of more sophisticated monitors like banks. Though several theories of trade credit have been put forth, the collateral theory—that suppliers place a higher liquidation value on collateral—is one prominent view. A critical assumption in this theory is that trade creditors have the legal right to the underlying collateral. This paper shows that in the presence of protective laws that support suppliers' ability to reclaim and liquidate their goods, trade credit can indeed have an important collateral component.

[35] It's difficult to directly compare the coefficients in Tables 4–9 since they include different samples and the dependent variables are scaled differently due to data restrictions.

BAPCPA provides a sharp, quasi-experimental setting to test the theories linking collateral to the extension of trade credit since it dramatically improved the trade creditor's ability to reclaim and liquidate the goods of bankrupt customers. The results show a powerful relation between an improvement in sellers' rights to collateral and their willingness to lend. However, I find that improvements in collateral protection reduce sellers' lending and monitoring standards, as customer portfolios become significantly riskier for the suppliers impacted by the legal change. The results mirror those shown in the securitization literature, particularly those that study the effect of MBS in the recent financial crisis.

This paper speaks to the broader literature on how legal institutions influence capital markets, which generally finds that stronger lender rights enhance the flow of debt capital (La Porta, Lopez-de Silanes, Shleifer and Vishny, 1997; 1998; Levine, 1998; 1999; Djankov, McLiesh and Shleifer, 2007). I show that a law that impacts one class of creditors within the US—trade creditors—improves their willingness to lend. However, by shifting collateral rights to trade creditors, other creditors face less protection and reduce lending. Thus, laws like BAPCPA that reallocate rights to some creditors can have important policy implications. If collateral is more valuable in the hands of trade creditors and market frictions impede the appropriate allocation of collateral, then intervention and strategic allocation by the courts may be favorable. However, if other lenders (i.e., banks) can more efficiently monitor collateral, then reallocation by the courts could have deleterious effects. While it's beyond the scope of this paper to analyze the welfare effects of BAPCPA, it's important to recognize that improving the rights of some creditors can have negative spillover effects that influence the behavior of other providers of capital. I leave it to future research to study the costs and benefits of laws like BAPCPA that target a subset of capital providers.

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